

Data 498D

Reflection Essay II

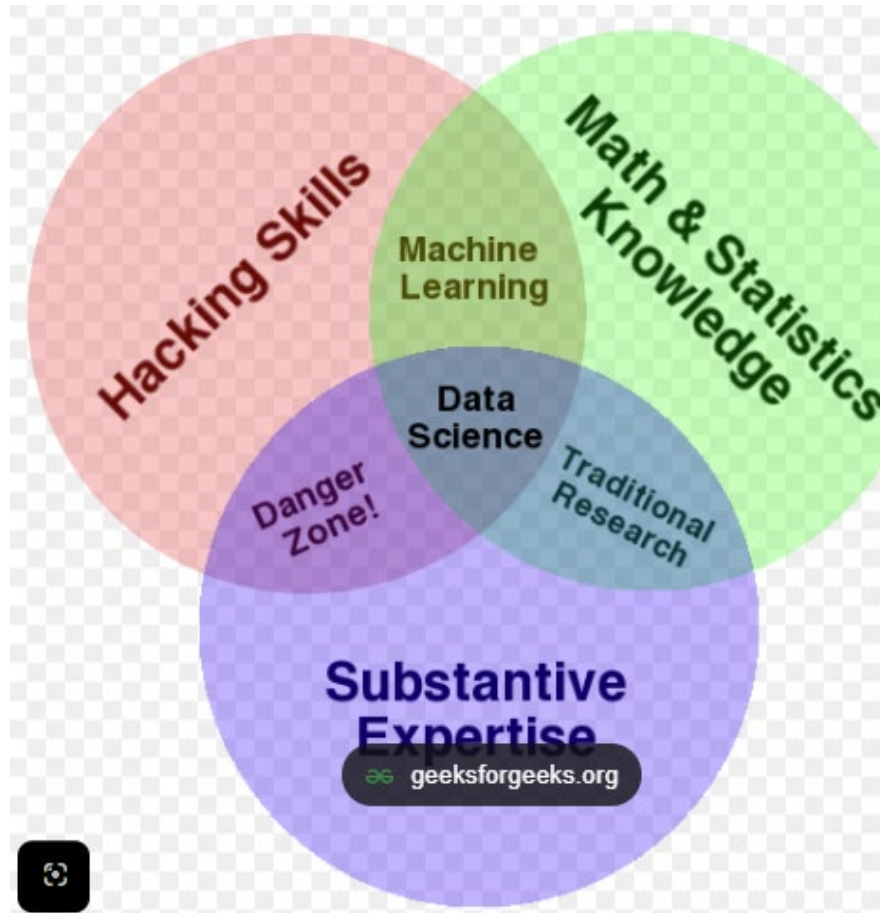
Joe Watkins & Lianfen Qian
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09/14/2026



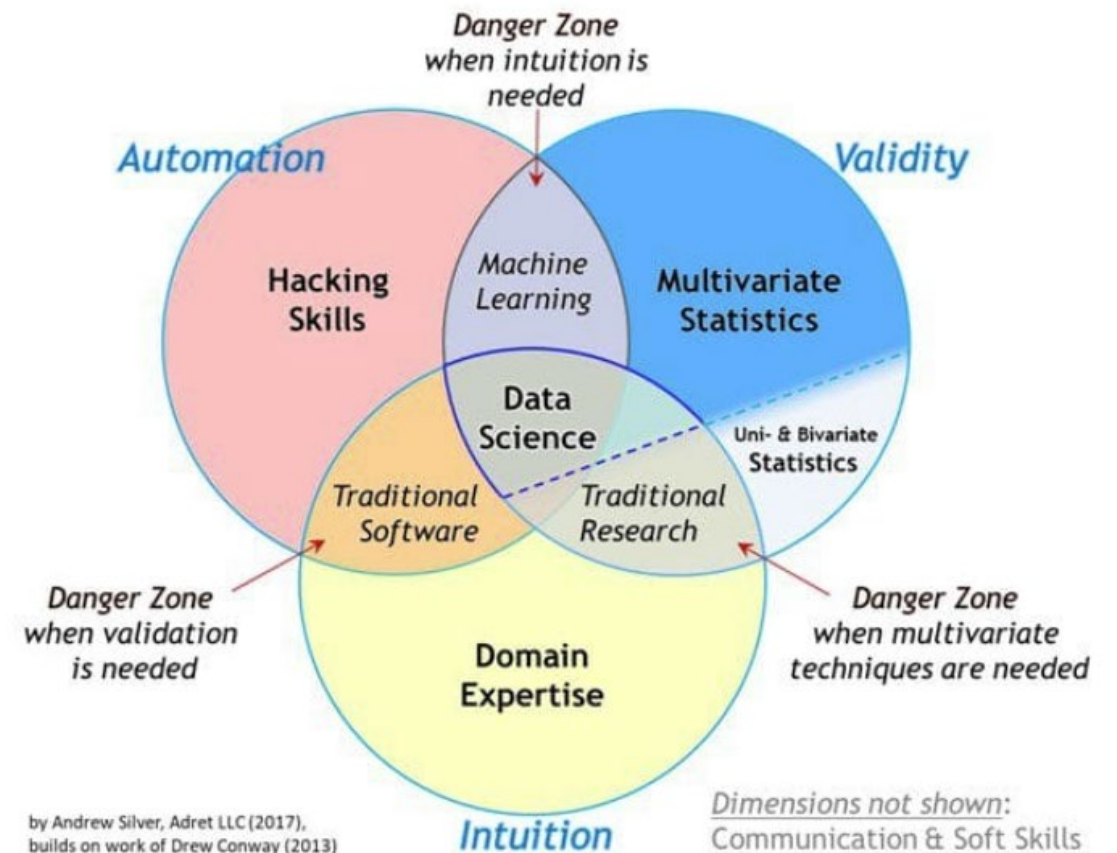
Goals for Today

- Motivation Example: Evolution of Data Science Life Cycle
- Gibbs' Reflective Cycle
- Reflective Essay Checklist
- Discuss the definitions of Data Science Life Cycle
- Present your talking points to a peer
- Response and Critique the talking points given by a peer
- **Wednesday:**
 - Introduce your selected scientific article to a peer
 - Present your talking points to a peer
 - Response and critique the talking points given by a peer

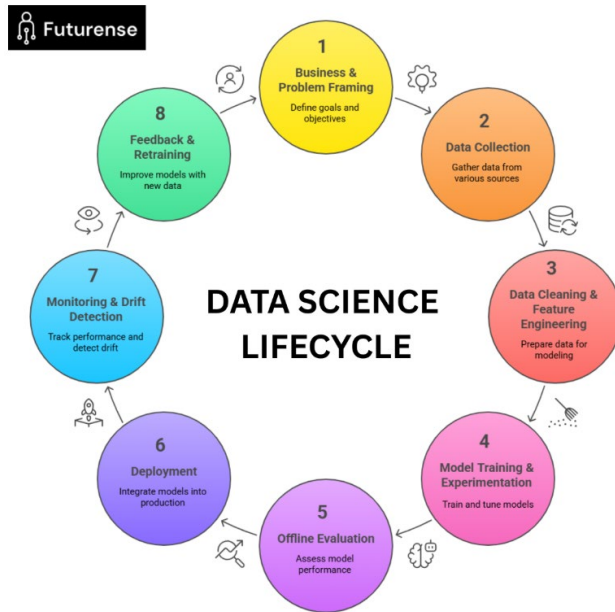
Drew Conway's Diagram of Data Science



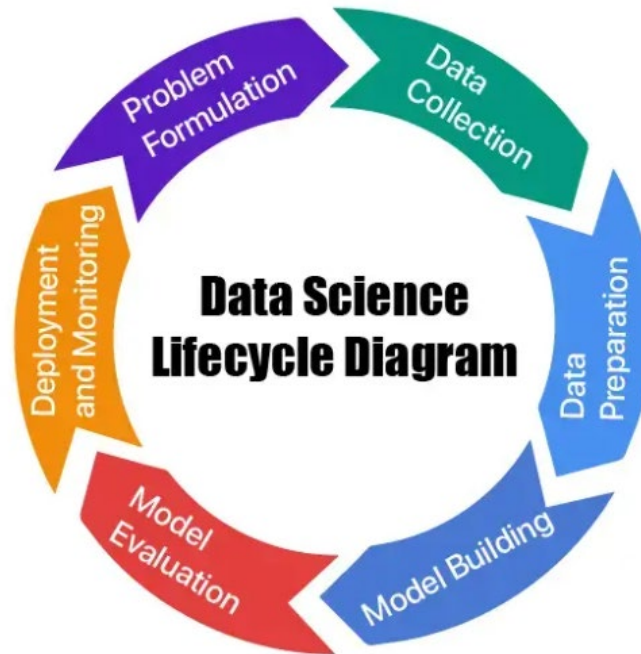
By Andrew Silver, Adret LLC



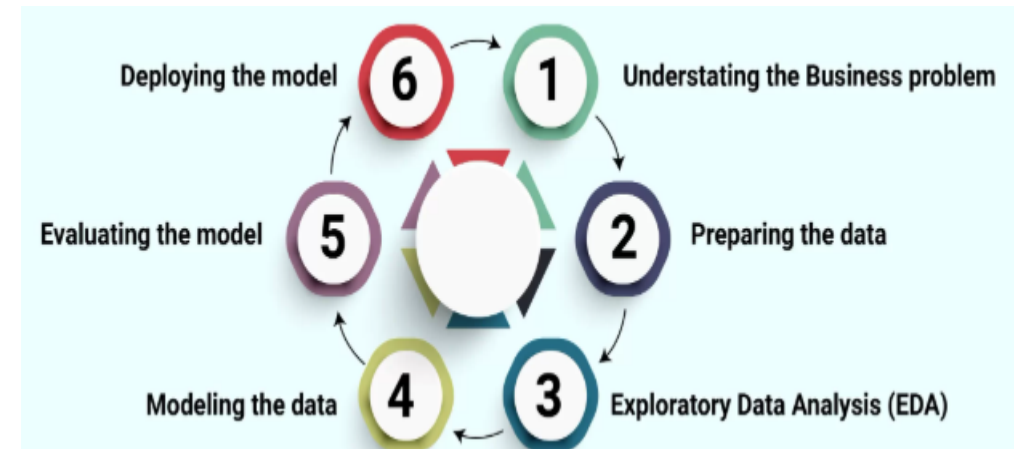
Data Science Life Cycle



<https://futureense.com/blog/data-science-process-with-diagram>

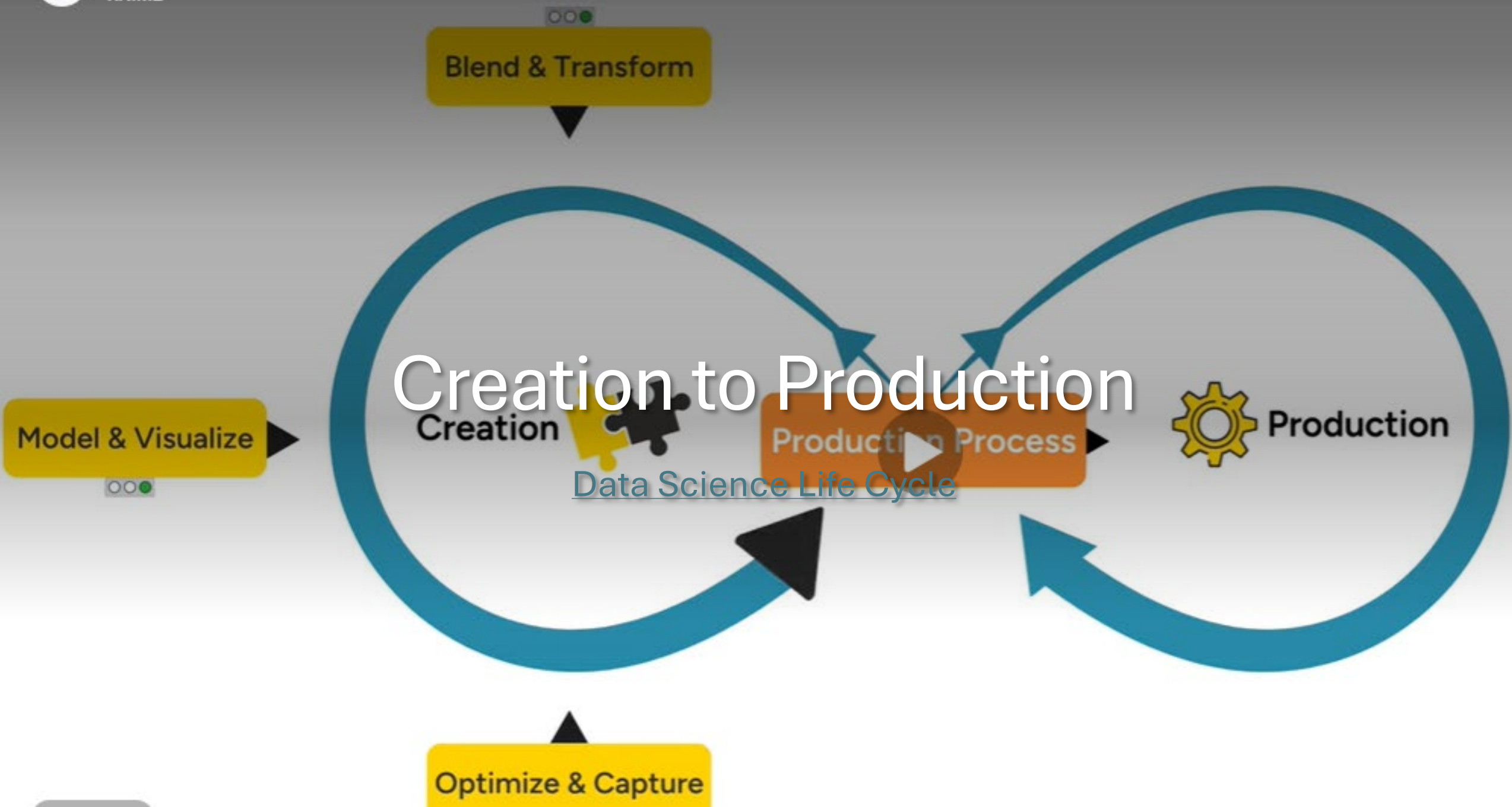


<https://www.upskillcampus.com/blog/data-science-lifecycle/>
https://www.tutorialspoint.com/data-science/data_science_lifecycle.htm

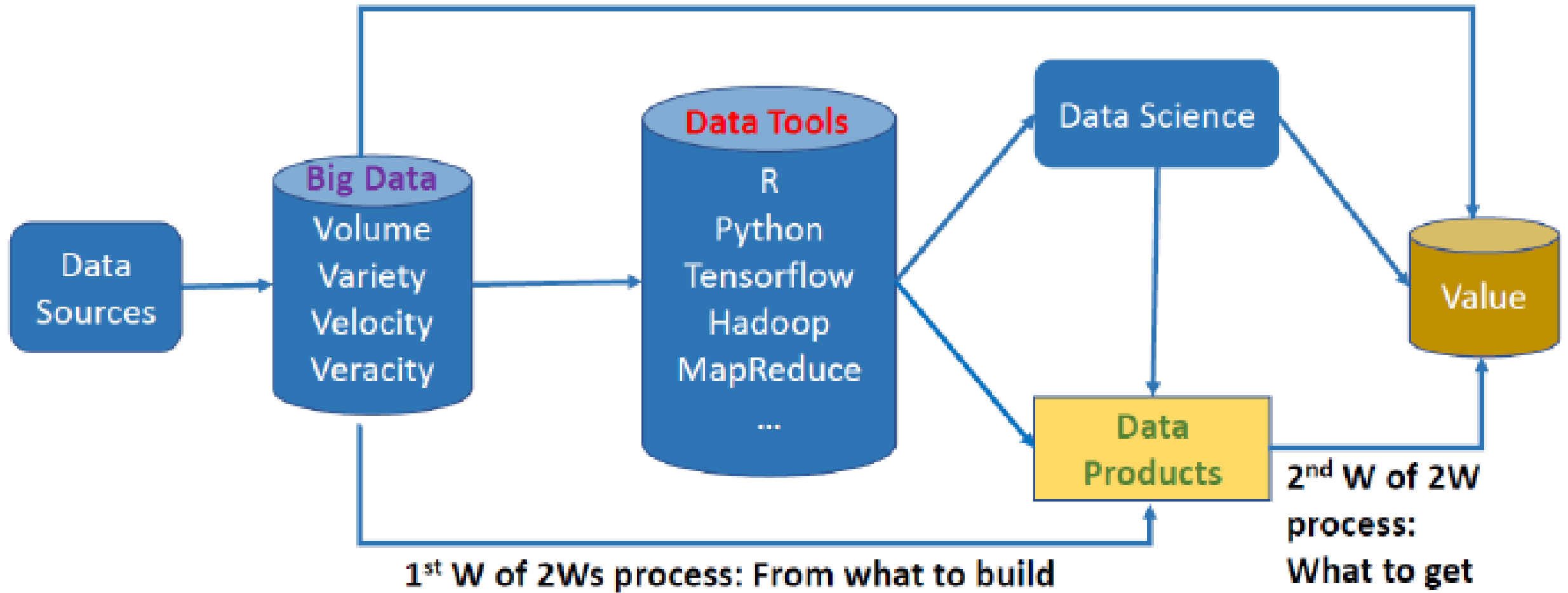


<https://www.theiotacademy.co/blog/data-science-lifecycle/>





From 4Vs to 1V

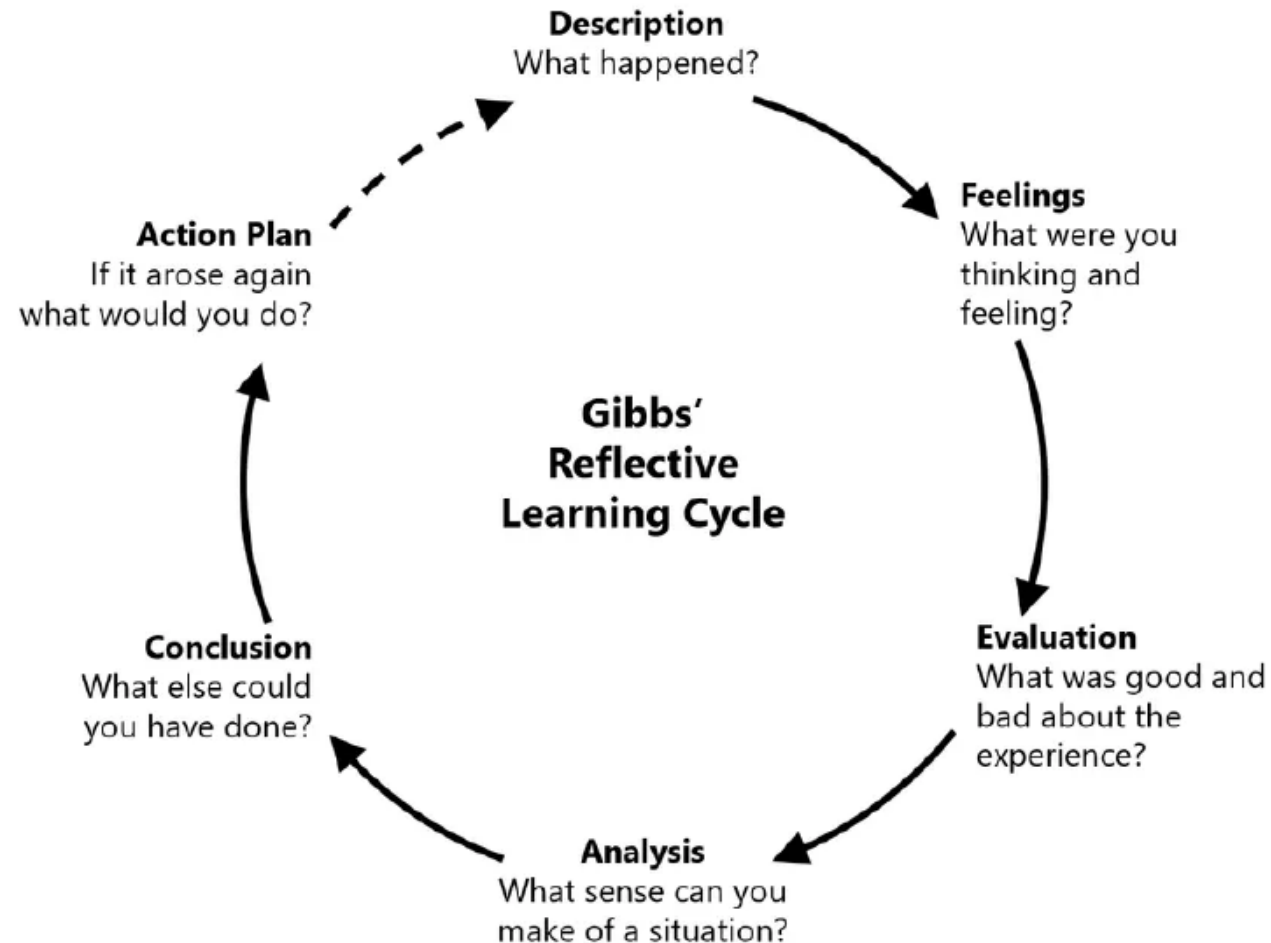


- Data Science Life Cycle

- **Problem Definition** – Every data science project begins by clearly defining the business problem or research question. For example, a retailer may want to predict customer churn.
- **Data Collection** – Relevant data is gathered from multiple sources such as databases, APIs, web scraping, or sensors. The quality and quantity of data directly affect the outcome.
-  **Data Preparation** – This involves cleaning missing values, handling duplicates, and transforming data into the right format for analysis. Data preparation often takes the most time in a project.
- **Exploratory  Data Analysis (EDA)** – Using statistical methods and visualization techniques,  data scientists identify patterns, correlations, and anomalies in the dataset.
- **Model Building** – Machine learning or statistical models are developed and trained using the prepared data. Techniques such as regression, classification, clustering, or deep learning may be used.
- **Model Evaluation** – Models are tested using metrics like accuracy, precision, recall, or F1-score to ensure they perform reliably before deployment.
- **Deployment** – The model is integrated into business systems or applications, where it begins generating predictions or automating decisions in real-time.
- **Monitoring & Maintenance** – Once deployed, models are continuously monitored and updated as new data becomes available to maintain accuracy and effectiveness.

Gibbs Reflective Cycle

Gibbs Reflective Cycle



Reflection Essay: A Simple Checklist

A strong reflection connects statistical understanding with critical reading

- ✓ What is the study trying to accomplish?
- ✓ What data were collected, and how were they analyzed?
- ✓ What did the authors conclude?
- ✓ How strong and generalizable are the claims?
- ✓ How do language, organization, graphs, and tables support the argument?
- ✓ What is your own critical assessment of the article?

Reflection Essay (5Ws):

- Introduction: What is the experience?
- Description: What happened?
- Reflection: What did you think and feel? (core)
- Analysis: What did you learn?
- Conclusion: What will you do differently?

A simple formula: What? → So what? → Now what?

A stronger version: Experience → Reflection → Analysis → Learning → Future Action

- Discuss:
 - What surprised you?
 - What was difficult?
 - What assumptions did you have?
 - Did your thinking change?
 - What mistakes did you make?
 - How did you respond to challenges?